

CSE 312 Midterm 1

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I. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

Given the following function

```

1 int algorithm1(int N){
2   int i, j, k, sum = 0;
3   for (i = 1; i <= N; i++)
4     for (j = i; j <= N - i; j++)
5       for (k = 1; k <= j - i; k++)
6         sum++;
7   return sum;
8 }
```

What is the time complexity of the function algorithm1?

II. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

Given the following function

```

1 int algorithm2(int N){
2   int sum = 0;
3   if (N == 0)
4     return 0;
5   for (i = 0; i < N; i++)
6     for (int j = 0; j < N; j++)
7       sum++;
8   return algorithm2(N - 1) + sum;
9 }
```

What is the time complexity of the function algorithm2?

III. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

- Prove that $\frac{1}{2}n(n-2) \in \Theta(n^2)$.
- Sort the following functions in the order of growth.

$2^n, n, n^3, n!, n \log n, n^2, \log n$

IV. QUESTION (BRUTE FORCE) (20 POINTS)

Write a brute force algorithm to calculate the least common multiple of two numbers m and n .

```
int lcm(int m, int n)
```

V. QUESTION (BRUTE FORCE) (20 POINTS)

You are given an integer matrix of $A[][]$, which is the adjacency matrix of a graph. Write a function that checks if the graph has the topology of a star.

```
boolean starTopology(int[][] A)
```

